

Clinical Risk Assessment Report: Patient 56

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.52x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.5%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Patient Age (Age)

Value: 48.0 [Bottom 7%] (+2.1%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)

Value: 0.7 [Top 24%] (+1.1%)

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

3. Protective / Mitigating Factors (Reducing Risk)

- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: -0.79 [Bottom 19%] (-3.5%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.
- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: 0.09 [Top 19%] (-1.4%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Minimum Tumor Density (CONVENTIONAL_HUmin)

Value: -2.01 [Bottom 3%] (-1.1%)

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.
- Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)

Value: 0.0 [Top 20%] (-0.9%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.
- Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)

Value: -0.18 [Avg (60th %ile)] (-0.9%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.
- Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)

Value: -0.42 [Bottom <1%] (-0.8%)

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)

